

H524 Introduction to Biostatistics

AI-Enhanced Public Health Data Analysis Project

Fall 2025

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Final Assessment Project Overview

Teams of 7 students will conduct a comprehensive biostatistical analysis of a real public health dataset as the culminating course assessment (25% of final grade). This project replaces traditional final exams and demonstrates effective integration of AI tools throughout the research process while maintaining scientific rigor and reproducibility.

1 Project Timeline

- **Week 3:** Project introduction and team formation
- **Week 4:** Dataset selection and research question formulation
- **Week 5:** Project proposal due (1 page)
- **Week 7:** Mid-project check-in and feedback
- **Week 9:** Draft analysis due for peer review
- **Week 10:** Final project submission
- **Finals Week:** Final assessment presentations (20 minutes + 5 minutes Q&A per team)

2 Available Datasets

Each team must select one of the following real public health datasets:

2.1 Option 1: CDC PLACES Health Data (County-Level)

- **Source:** CDC PLACES Project 2022
- **Focus Areas:** Chronic disease prevalence, health behaviors, prevention measures
- **Sample Size:** 3,100 counties/county equivalents
- **Access:** <https://chronicdata.cdc.gov/browse?category=500+Cities+%26+Places>
- **Data Type:** Aggregated county-level estimates (no survey weights needed)

- **Suggested Research Questions:**

- Geographic patterns in diabetes prevalence across US counties
- Association between preventive care access and health outcomes
- Rural vs urban differences in chronic disease burden

2.2 Option 2: COVID-19 Case Surveillance Public Use Data

- **Source:** CDC COVID-19 Case Surveillance
- **Focus Areas:** COVID-19 outcomes, risk factors, demographics
- **Sample Size:** 1 million de-identified cases
- **Access:** <https://data.cdc.gov/Case-Surveillance/>
- **Suggested Research Questions:**
 - Risk factors for severe COVID-19 outcomes
 - Geographic and temporal patterns in case fatality rates
 - Age-stratified analysis of hospitalization predictors

2.3 Option 3: Heart Disease Mortality Data (County-Level)

- **Source:** CDC WONDER Underlying Cause of Death
- **Focus Areas:** Heart disease mortality rates, demographic patterns
- **Sample Size:** 3,100 counties with 5+ years of data
- **Access:** <https://wonder.cdc.gov/controller/datarequest/D77>
- **Data Type:** Aggregated mortality rates (no survey weights needed)
- **Suggested Research Questions:**
 - Geographic disparities in heart disease mortality
 - Age and gender patterns in cardiovascular death rates
 - Trends in heart disease mortality over time

2.4 Option 4: Global Burden of Disease (GBD) Study

- **Source:** IHME GBD 2019
- **Focus Areas:** Disease burden, mortality, risk factors globally
- **Sample Size:** 204 countries and territories
- **Access:** <http://ghdx.healthdata.org/gbd-results-tool>
- **Suggested Research Questions:**
 - Trends in disease burden across income levels
 - Environmental risk factors and respiratory disease mortality
 - Progress toward health-related SDG targets

2.5 Option 5: Youth Tobacco Use Data

- **Source:** CDC Youth Risk Behavior Surveillance System (State-Level)
- **Focus Areas:** Tobacco use patterns, demographics, risk behaviors
- **Sample Size:** 50 states + DC, multiple years
- **Access:** <https://www.cdc.gov/healthyyouth/data/yrbs/>
- **Data Type:** State-level aggregated percentages (no survey weights needed)
- **Suggested Research Questions:**
 - State-level variations in youth tobacco use
 - Trends in e-cigarette vs traditional cigarette use
 - Association between tobacco policies and use rates

2.6 Option 6: OpenFDA Adverse Events Database

- **Source:** FDA Adverse Event Reporting System (FAERS)
- **Focus Areas:** Drug safety, adverse events, pharmacovigilance
- **Sample Size:** Millions of reports
- **Access:** <https://open.fda.gov/data/faers/>
- **Suggested Research Questions:**
 - Patterns in adverse events for specific drug classes
 - Age and gender differences in drug safety profiles
 - Time trends in reporting of serious adverse events

3 Project Components

3.1 Part 1: Research Question and Literature Review (15%)

- Formulate a clear, answerable research question
- Conduct literature review using AI-assisted search strategies
- Document AI tool usage for literature synthesis
- Identify gaps your analysis will address

3.2 Part 2: Data Management and Exploration (20%)

- Data cleaning and preprocessing with AI assistance
- Exploratory data analysis with appropriate visualizations
- Missing data assessment and handling strategy
- Document all AI-generated code with verification steps

3.3 Part 3: Statistical Analysis (30%)

- Apply appropriate statistical methods from H524 curriculum:
 - Descriptive statistics and data visualization
 - Hypothesis testing (t-tests, chi-square, ANOVA)
 - Correlation and regression analysis
 - Appropriate adjustments for multiple comparisons
- Use AI to explore alternative analytical approaches
- Validate all AI-suggested methods against course materials
- Conduct sensitivity analyses where appropriate

3.4 Part 4: AI Integration and Documentation (20%)

- Maintain an AI usage log throughout the project
- Document all prompts used for substantial tasks
- Include verification steps for AI-generated results
- Reflect on AI benefits and limitations encountered
- Create a "lessons learned" section on AI use

3.5 Part 5: Final Presentation and Communication (25%)

- 20-minute team presentation + 5 minutes Q&A
- Clear visual aids and professional data visualizations
- Effective storytelling with data and public health implications
- Demonstration of team collaboration and individual contributions
- Professional delivery, time management, and audience engagement

4 Deliverables

4.1 Project Proposal (Week 5)

One-page proposal including:

- Research question and hypothesis
- Selected dataset and justification
- Planned statistical approaches
- Team member roles and responsibilities
- Anticipated AI tools and usage

4.2 Final Report (Week 10)

Comprehensive report (12-15 pages) including:

- Executive summary (1 page)
- Introduction and literature review
- Methods section with data description
- Results with tables and figures
- Discussion and limitations
- AI usage documentation appendix
- Reproducible R code with comments
- Team contribution statement

4.3 Final Assessment Presentation Materials (Finals Week)

- PowerPoint/PDF slides (20-25 slides for 20-minute presentation)
- Live demonstration of key findings and statistical methods
- Prepared responses for anticipated questions and statistical challenges
- One-page executive summary handout for audience
- Individual reflection on learning and AI tool usage (1 page per student)

5 AI Usage Requirements

Required AI Integration:

1. **GitHub Copilot:** Use for R code generation and debugging
2. **Literature Review:** Use AI to identify relevant papers and synthesize findings
3. **Statistical Methods:** Consult AI for method selection and interpretation
4. **Data Visualization:** Generate initial visualization code with AI assistance
5. **Writing Support:** Use AI for editing and clarity (with attribution)

AI Documentation Template:

```
## AI Usage Log Entry
Date: [Date]
Tool: [GitHub Copilot/ChatGPT/Claude/etc.]
Task: [Specific task]
Prompt: [Exact prompt used]
Output: [Summary of AI response]
Verification: [How you verified the output]
Modifications: [Changes made to AI output]
```

6 Grading Rubric

Component	Weight	Points
Research Question & Literature Review	15%	15
Data Management & Exploration	20%	20
Statistical Analysis	25%	25
AI Integration & Documentation	15%	15
Final Presentation & Communication	25%	25
Total	100%	100

6.1 Detailed Grading Criteria

6.1.1 Excellent (90-100%)

- Clear, innovative research question with public health relevance
- Comprehensive statistical analysis with appropriate methods
- Exceptional AI integration with thorough documentation
- Professional presentation with compelling data story
- All team members contributed meaningfully

6.1.2 Good (80-89%)

- Well-defined research question with clear objectives
- Appropriate statistical methods correctly applied
- Good AI usage with adequate documentation
- Clear presentation with good visualizations
- Most team members actively participated

6.1.3 Satisfactory (70-79%)

- Basic research question addressed
- Statistical methods mostly correct with minor errors
- AI used but documentation incomplete
- Adequate presentation meeting time requirements
- Uneven team contribution evident

6.1.4 Needs Improvement (<70%)

- Unclear research question or objectives
- Inappropriate statistical methods or major errors
- Minimal AI usage or poor documentation
- Presentation lacks clarity or professionalism
- Limited team collaboration

7 Team Formation and Roles

Each team of 7 students should designate:

- **Project Manager:** Coordinate meetings, deadlines, and submissions
- **Data Manager:** Lead data cleaning and preprocessing
- **Statistical Lead:** Guide analytical approach and methods
- **AI Integration Lead:** Coordinate AI tool usage and documentation
- **Visualization Lead:** Create figures and presentation materials
- **Writing Lead:** Coordinate report writing and editing
- **Presentation Lead:** Coordinate final presentation and Q&A preparation

Note: All team members are expected to contribute to all aspects; roles are for coordination.

8 Resources and Support

8.1 Technical Resources

- RStudio with GitHub Copilot enabled
- Access to OSU library databases
- Canvas discussion board for project Q&A
- Sample analysis templates on Canvas

8.2 Office Hours Support

- Regular office hours: Wednesday 10:00-11:30 AM
- Special project office hours: Weeks 5, 7, 9
- Schedule team consultations via Canvas

8.3 Recommended R Packages

```
# Data manipulation
library(tidyverse)
library(data.table)

# Statistical analysis
library(stats)
library(car)
library(lmtest)
library(tableone)

# Visualization
library(ggplot2)
library(plotly)
library(corrplot)

# Reporting
library(knitr)
library(rmarkdown)
library(stargazer)
```


9 Academic Integrity

Important Reminders:

- All AI-generated content must be verified and attributed
- Each team member must contribute substantively
- Plagiarism or fabrication of data will result in project failure
- Inter-team collaboration is encouraged, but each team's work must be unique
- Citation of all sources (including AI tools) is required

10 Tips for Success

1. **Start Early:** Data cleaning always takes longer than expected
2. **Communicate Regularly:** Schedule weekly team meetings
3. **Document Everything:** Keep detailed notes of decisions and methods
4. **Verify AI Output:** Always double-check AI-generated code and interpretations
5. **Practice Presentation:** Rehearse timing and transitions
6. **Seek Feedback:** Use office hours and peer review opportunities
7. **Focus on Story:** Ensure your analysis tells a coherent public health story